Pass Network Analysis of FIFA World Cup 2026 Final Match

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Abstract

This paper presents a Social Network Analysis of the 2026 FIFA World Cup final between Spain and Argentina, held on July 19, 2026, in which Spain won with a single goal scored by Ferran Torres. The analysis models each team's passing behavior as a directed weighted graph where nodes represent players positioned at their real average pitch coordinates and edges represent completed passes weighted by frequency. Using centrality measures, temporal phase decomposition, Louvain community detection, and directed triangle analysis, the study demonstrates that Spain's structural superiority in ball control, passing density, and tactical organization was the decisive factor in the match outcome. Spain recorded 792 completed passes against Argentina's 377, achieved a network density of 0.599, and identified three cohesive tactical communities whose coordinated interaction enabled the winning goal. Argentina's passing network degraded significantly following the red card issued to Enzo Fernández at expanded minute 97, leaving them with 10 players through extra time and virtually no ball possession in the immediate aftermath of the dismissal.

Index Terms

Social Network Analysis, Pass Networks, Graph Theory, Centrality Measures, Community Detection, Football Analytics, FIFA World Cup 2026

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I. Introduction

The 2026 FIFA World Cup final between Spain and Argentina was one of the most debated matches in recent football history. Spain dominated possession from the opening minutes, prompting widespread discussion about whether Argentina's compact defensive strategy was a deliberate tactical choice or simply an inability to compete with Spain's ball circulation. Detractors of the result argued that the outcome reflected external circumstances rather than footballing merit, while Spain's supporters pointed to the sheer volume of passing sequences and territorial control as evidence of structural superiority.

This paper approaches that debate through the lens of graph theory and social network analysis. By modeling each team's passing behavior as a directed weighted network, it becomes possible to quantify claims about ball dominance, identify which players were structurally indispensable, detect the tactical communities that formed around the winning goal, and measure how Argentina's network degraded after their red card. The analysis does not rely on subjective observation but on computable graph properties including centrality measures, clustering coefficients, path lengths, and directed triangle frequencies.

From a practical standpoint, this type of analysis has direct applications for coaching staff at professional clubs. Identifying which players act as structural bridges, which triangular combinations are most reliable under pressure, and how a team's passing network evolves across match phases gives a data-driven foundation for tactical preparation, substitution decisions, and opposition scouting. The methodology presented here is generalizable to any match dataset that includes pitch coordinates and event timestamps, making it a reusable tool for football analytics.

II. Theoretical Framework

A graph $G = (V, E)$ is a mathematical structure composed of a set of vertices V and a set of edges E that connect pairs of vertices [1]. When edges carry a direction, the graph is called directed or a digraph; when they additionally carry a numerical value, the graph is called weighted. In this analysis, the pass network of each team is modeled as a directed weighted graph where V is the set of players and each edge (u, v, w) represents the fact that player u passed to player v a total of w times during the match. The weight w is an integer greater than zero, since only completed passes are included.

The adjacency matrix A of a directed graph is a square matrix of size $|V| \times |V|$ where $A_{ij} = w$ if a directed edge from node i to node j exists with weight w, and $A_{ij} = 0$ otherwise. For a directed network, A is generally asymmetric, since a pass from player i to player j does not imply the same frequency in the reverse direction [1]. The degree of a node in a directed graph is split into in-degree, the number of incoming edges, and out-degree, the number of outgoing edges. A directed cycle is a closed path $v_1 \to v_2 \to \cdots \to v_k \to v_1$ where all edges follow their directed orientation, which in the context of football corresponds to a passing triangle when $k = 3$.

A network's size is defined by its number of nodes N and edges L. The density of a directed network is given by:

$$\rho = \frac{L}{N(N-1)} \tag{1}$$

and measures how close the network is to being fully connected [2]. The average shortest path length $\langle d \rangle$ is the mean of the shortest path distances between all pairs of nodes, and serves as a measure of how efficiently information or the ball can travel through the network. The diameter d_{max} is the longest shortest path in the network, and the radius r is the minimum eccentricity across all nodes, where the eccentricity of node i is $\epsilon(i) = \max_j d(i, j)$. Nodes with eccentricity equal to the radius form the center of the graph, while those with eccentricity equal to the diameter form the periphery. The clustering coefficient of a node i measures the fraction of its neighbors that are also connected to each other, and the average clustering coefficient $\langle C \rangle$ averages this over all nodes [2].

Centrality measures quantify the structural importance of a node within a network [3]. In-degree centrality counts the normalized number of incoming connections and identifies the most sought-out receivers. Out-degree centrality counts outgoing connections and identifies the most active distributors. Betweenness centrality measures the fraction of all shortest paths in the network that pass through a given node:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where σ_{st} is the total number of shortest paths from s to t and $\sigma_{st}(v)$ is the number of those paths that pass through v [3]. A high betweenness node is a structural bridge whose removal would fragment the network. Eigenvector centrality assigns importance to a node proportionally to the importance of its neighbors, capturing the notion of synergy within a team. PageRank extends this idea to directed networks with a damping factor that models a random walk through the graph, making it particularly appropriate for weighted directed pass networks where the direction of the pass matters [2].

In large social networks, degree distributions often follow a power law, a signature of scale-free networks generated by preferential attachment [4]. Pass networks of 17 players do not follow this pattern because the system is bounded and tactically constrained: every player must both send and receive passes within a formation. The degree distribution in a pass network is therefore more uniform, and any skew reveals tactical concentration. Beyond degree, the frequency of directed triangles (motifs of the form $A \to B \to C \to A$) is a key indicator of possession-based play. Research on large football datasets has shown that teams with higher triangle motif frequencies tend to sustain longer passing sequences and maintain greater territorial control [5].

Community detection identifies groups of nodes that interact more densely among themselves than with the rest of the network. The Louvain method [6] is a hierarchical modularity optimization algorithm that iteratively assigns nodes to communities by maximizing the modularity score:

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{3}$$

where m is the total number of edges, k_i is the degree of node i, c_i is the community assignment of node i, and δ is the Kronecker delta. Compared to hierarchical agglomerative methods, which require computing a full distance matrix, Louvain scales efficiently to large networks while still producing interpretable community structures.

A football pass network is a natural application of directed weighted graphs because passes have a sender, a receiver, and a frequency [7]. Unlike social networks where edges represent abstract relationships, pass network edges are directly observable events with spatial and temporal attributes. The in-degree of a player corresponds to how reliably they receive the ball without losing possession. The betweenness centrality identifies the player whose absence would most disrupt the team's ability to circulate the ball. PageRank identifies the player who most consistently receives passes from other important players, which in football terms is closest to identifying the player with the highest overall ball dominance. Communities in a pass network map onto tactical subunits, lines of the formation or combination-play groups, whose internal cohesion is measurable and comparable between teams [7].

III. Methodology

A. Network Construction

The raw data for this analysis is a match event dataset for the 2026 FIFA World Cup final, collected and curated by Noah Bair and sourced from WhoScored [8]. The dataset contains one row per match event, with columns for player name, team, event type, outcome, pitch coordinates x and y (ranging from 0 to 100), expanded minute, period name, and a boolean flag indicating whether the event constituted a touch. The construction of the pass graphs was implemented in a dedicated Jupyter notebook using NetworkX and pandas.

Only completed passes (event type equal to Pass and outcome equal to Successful) were included. The receiver of each pass was inferred as the next player from the same team to register a touch event in chronological order after the pass, without crossing period boundaries. This inference strategy is robust for event-level data where the receiver is not explicitly encoded. Each player's average pitch position was computed as the mean of all their touch coordinates across the entire match, providing a position that reflects their actual area of operation rather than a nominal formation assignment. The resulting graphs were exported as weighted edgelist files in a custom format that encodes source, target, weight, and average position coordinates for both endpoints. Separate edgelist files were produced for each team at full-match level and for each of four temporal phases: first half, second half before the red card, second half after the red card, and extra time.

B. Network Analysis

The analysis notebook loaded the edgelist files and computed network characteristics, centrality measures, degree distributions, and community structure for both teams. All metrics were computed on the full 17-node graph since the small network size made exact computation feasible without sampling. Distance metrics including diameter, radius, eccentricity, periphery, and center were computed on the largest weakly connected component of each graph, which in both cases encompassed all 17 players. Temporal phase analysis used the phase-level edgelist files to track how each team's passing structure evolved across the match, with particular attention to the impact of the red card issued to Enzo Fernández at expanded minute 97. Community detection was applied using the Louvain algorithm on the undirected version of each team's full-match graph, with a fixed random seed of 42 for reproducibility. Directed triangle detection enumerated all cycles of length 3 in the directed graph and ranked them by bottleneck weight, defined as the minimum edge weight among the three edges of the triangle. The primary libraries used were NetworkX for graph operations, python-louvain for community detection, and matplotlib and seaborn for visualization.

IV. Results and Discussion

A. Network Classification

Both networks are directed, weighted, cyclic, and non-bipartite. They are not planar given the density of crossing edges visible in the visualizations. Each team's graph forms a single weakly connected component and a single strongly connected component, meaning every player can reach every other player through directed pass sequences, and the ball can circulate to any part of the formation without structural dead ends.

Node positions in Fig. 1 and Fig. 2 are not assigned by formation label but computed as the average of each player's real pitch coordinates across every touch in the match. This makes the spatial layout analytically meaningful. Spain's players are distributed across the full extent of the pitch, with attacking and midfield players positioned well into the opponent's half. Argentina's players, by contrast, are clustered predominantly in their own half, with only Messi, Giuliano Simeone, and Nico González making occasional excursions toward Spain's goal. This spatial distribution is a direct visual confirmation of Spain's territorial dominance throughout the match.

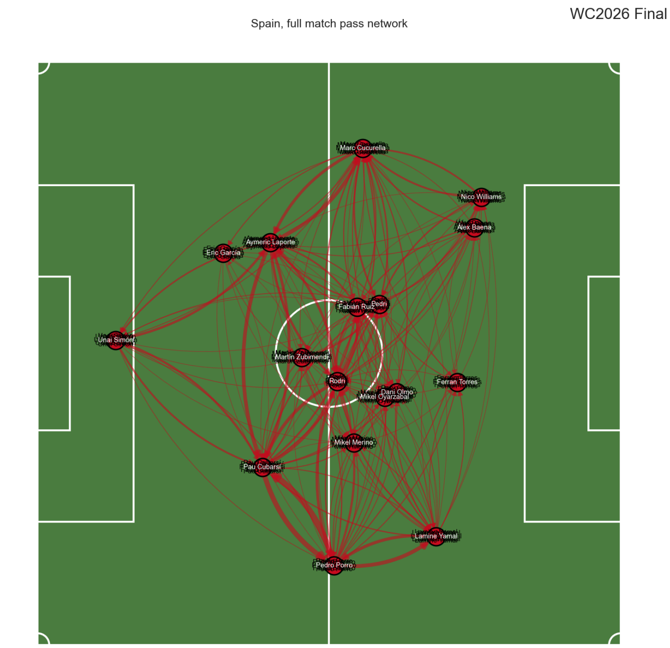

Fig. 1. Spain full-match pass network on pitch background. Node positions reflect real average pitch coordinates. Edge thickness is proportional to pass frequency.

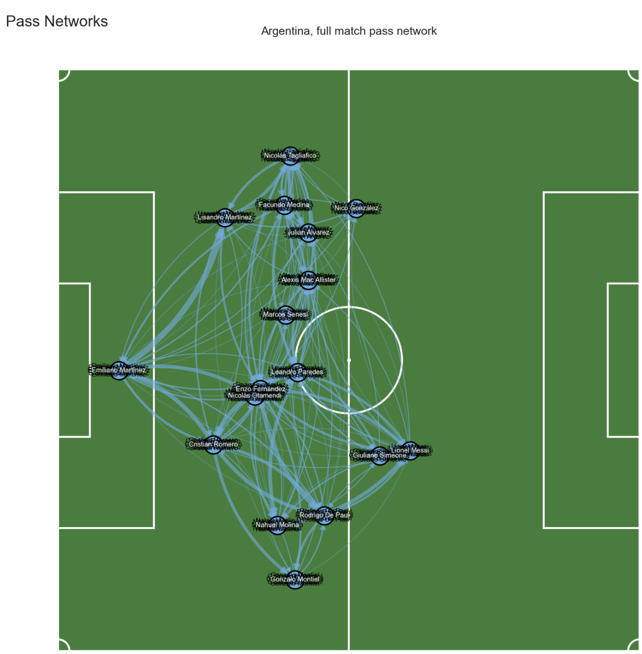

Fig. 2. Argentina full-match pass network on pitch background. The network is concentrated in Argentina's own half, reflecting their defensive posture throughout the match.

B. Network Characteristics

Table I presents the basic structural metrics for both networks. Spain's pass count of 792 is more than double Argentina's 377, establishing a quantitative basis for the widely observed perception of Spanish dominance. The density of Spain's network (0.599) is higher than Argentina's (0.540), indicating that a greater fraction of all possible player-to-player passing combinations were actually used during the match.

TABLE I
BASIC NETWORK CHARACTERISTICS: SPAIN VS ARGENTINA

Metric	Spain	Argentina
Number of nodes (players)	17	17
Number of edges (unique connections)	163	147
Total passes (sum of weights)	792	377
Density	0.5993	0.5404
Weakly connected components	1	1
Strongly connected components	1	1
Avg. clustering coefficient	0.0748	0.2255
Transitivity	0.7641	0.7306
Avg. shortest path length	1.4044	1.4853
Diameter	3	3
Radius	2	2

The average clustering coefficient reveals an interesting contrast. Argentina's coefficient (0.225) is nearly three times higher than Spain's (0.075). On the surface this might suggest Argentina's players were more locally cohesive, but in the context of a match where Argentina had far fewer passes, a higher clustering coefficient reflects the formation of small, tight passing clusters concentrated near their own penalty area, a symptom of defensive compression rather than an indicator of tactical sophistication. Spain's lower clustering coefficient is consistent with a more distributed possession style where the ball circulates across the entire width and depth of the pitch rather than cycling through a small group of players.

The average shortest path length (1.404 for Spain, 1.485 for Argentina) and identical diameter and radius values (3 and 2 respectively) indicate that both networks have comparable information flow efficiency. The center of Spain's graph encompasses 16 of 17 players, with only Mikel Oyarzabal on the periphery. Argentina's center includes 10 players while 7 players, predominantly defensive, sit on the periphery with eccentricity 3. This confirms that Argentina's peripheral players had limited passing reach into the broader team structure.

C. Centrality Measures

Table II summarizes the top-ranked player per team for each centrality measure.

TABLE II
TOP PLAYER PER CENTRALITY MEASURE

Measure	Spain	Argentina
In-degree	Marc Cucurella (0.938)	Enzo Fernández (0.750)
Out-degree	Pedro Porro (0.938)	Emiliano Martínez (0.750)
Betweenness	Marc Cucurella (0.097)	Enzo Fernández (0.076)
Eigenvector	Marc Cucurella (0.343)	Enzo Fernández (0.318)
PageRank	Pau Cubarśi (0.111)	Enzo Fernández (0.092)

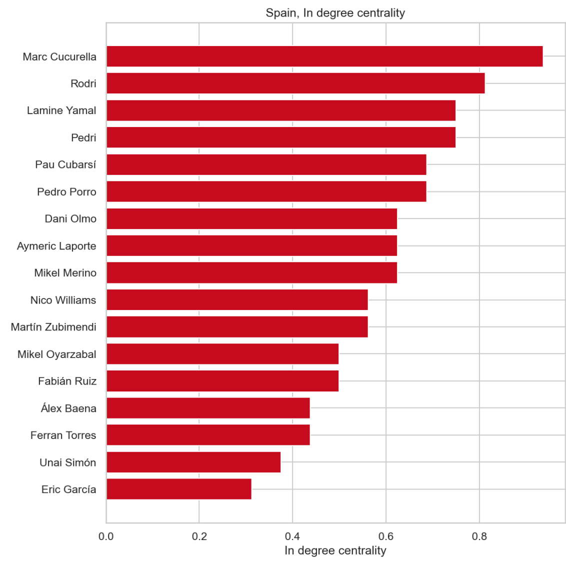

Fig. 3. In-degree centrality for Spain. Marc Cucurella leads as the most sought-out receiver.

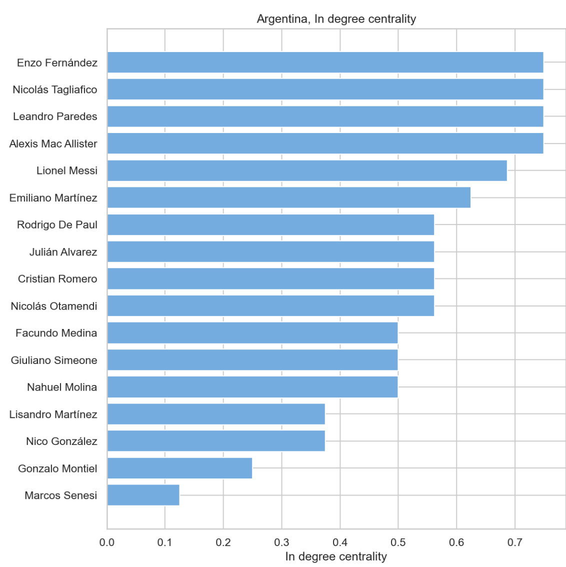

Fig. 4. In-degree centrality for Argentina. Enzo Fernández, Leandro Paredes, and Nicolás Tagliafico share the top position.

In-degree centrality, interpreted as the ability to receive passes successfully without losing the ball, places Marc Cucurella at the top of Spain's rankings. In football terms, a high in-degree player is one whom teammates trust to receive the ball in difficult situations. Pedro Porro's leadership in out-degree (0.938) identifies him as Spain's most active distributor, the player who most frequently initiated passing sequences toward other teammates.

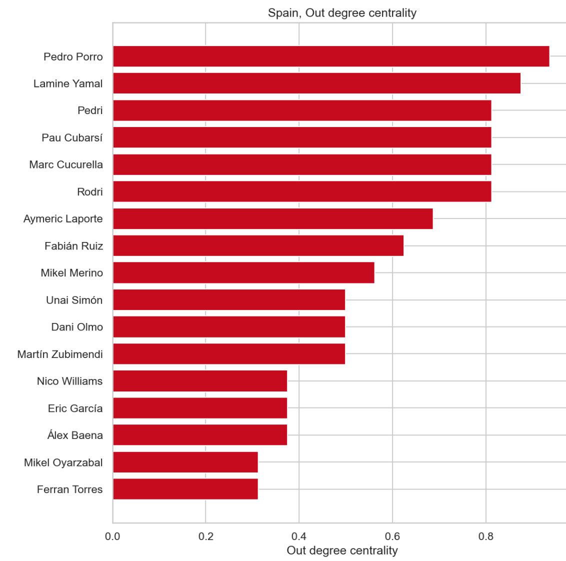

Fig. 5. Out-degree centrality for Spain. Pedro Porro was the most active pass distributor.

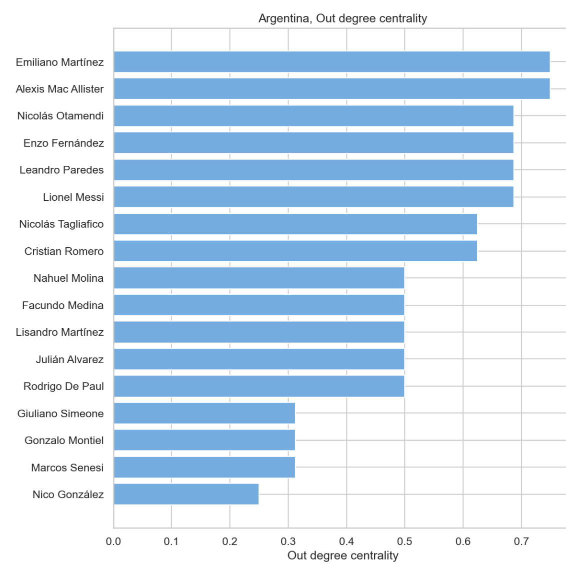

Fig. 6. Out-degree centrality for Argentina. Emiliano Martínez and Alexis Mac Allister lead in pass distribution.

Betweenness centrality presents one of the most structurally significant findings of this analysis. In the majority of football matches, the player with the highest betweenness centrality is a central midfielder, reflecting the traditional role of the pivot as the connective tissue of the team. In this match, that role was occupied by Marc Cucurella, a left back. This is a notable deviation from the expected pattern and suggests that Spain's tactical structure deliberately routed ball circulation through the left defensive channel, making Cucurella the indispensable bridge between Spain's defensive line and attacking units. His removal from the game would have been more disruptive to Spain's passing network than the removal of any midfielder.

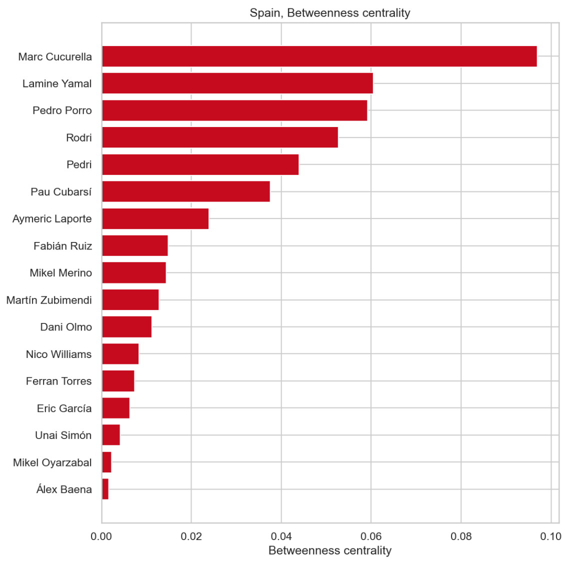

Fig. 7. Betweenness centrality for Spain. Marc Cucurella, a left back, leads ahead of all midfielders.

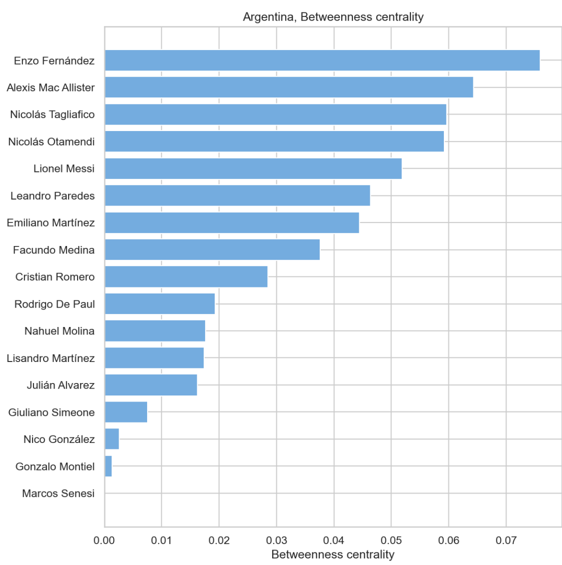

Fig. 8. Betweenness centrality for Argentina. Enzo Fernández leads, consistent with his central midfield role.

PageRank, the most appropriate centrality measure for weighted directed networks, can be interpreted intuitively as follows: if a spectator were to pause a replay of Spain's passing sequences at a random moment, PageRank gives the probability that each player would be the one with the ball at that instant, weighted by the importance of who passed it to them. Pau Cubarśi leads this metric for Spain (0.111), ahead of Marc Cucurella and Pedro Porro. This result reflects a structural reality of the match: Argentina's defensive attention was disproportionately focused on Lamine Yamal, whom they considered the most dangerous Spanish attacker. Yamal was frequently forced to return the ball to Porro, Rodri, or directly to Cubarśi. Simultaneously, Cubarśi served as the lateral relay point connecting Spain's left side (Laporte, Cucurella) to the right side (Porro), making him the node that the ball most frequently had to pass through when Spain was transitioning between flanks. Combined with Cubarśi's effectiveness in recovering the ball from Argentine forwards, these factors explain why he accumulates the highest PageRank despite not being the team's primary creative player.

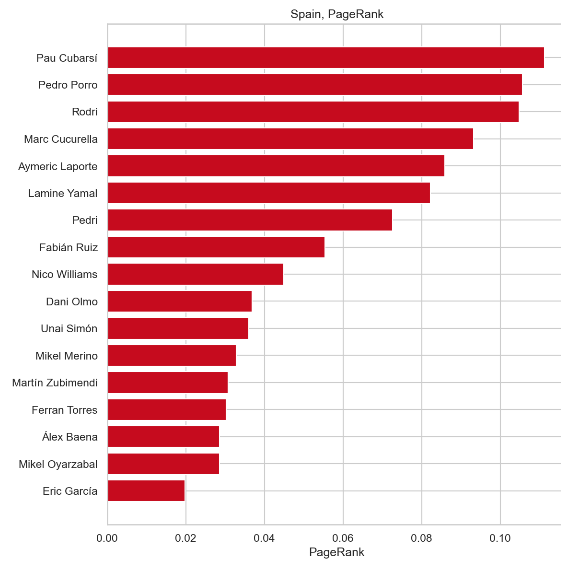

Fig. 9. PageRank for Spain. Pau Cubarśi leads, reflecting his role as the central relay node in Spain's circulation pattern.

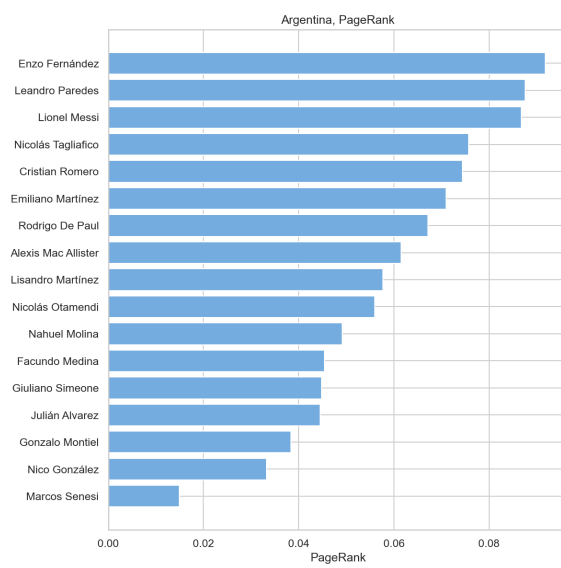

Fig. 10. PageRank for Argentina. Enzo Fernández leads, underscoring the structural impact of his red card on Argentina's network.

Eigenvector centrality measures which player generated the most synergy with their teammates, not merely in terms of pass volume but in terms of elevating the connectivity of the players around them. Marc Cucurella leads this metric for Spain (0.343). His passing combinations with Ferrán Torres, Fabián Ruiz, Rodri, Oyarzabal, Pedri, Dani Olmo, and Zubimendi were the connective thread that held Spain's attacking structure together. Crucially, Cucurella's circular passing sequences through Pedri and Nico Williams created the spatial conditions for the only goal of the match: a combination that ended with Nico Williams delivering the assist to Ferrán Torres, who finished centrally.

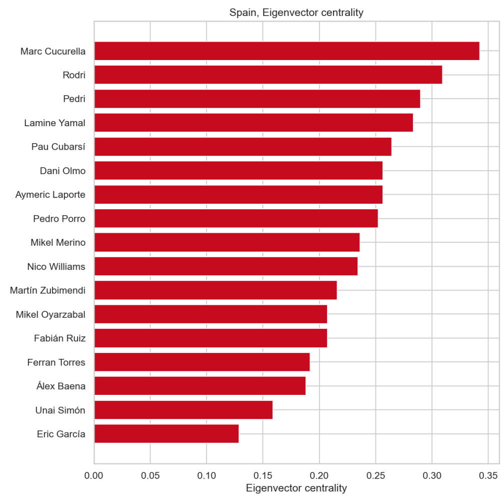

Fig. 11. Eigenvector centrality for Spain. Marc Cucurella leads, reflecting the synergistic impact of his play on surrounding teammates.

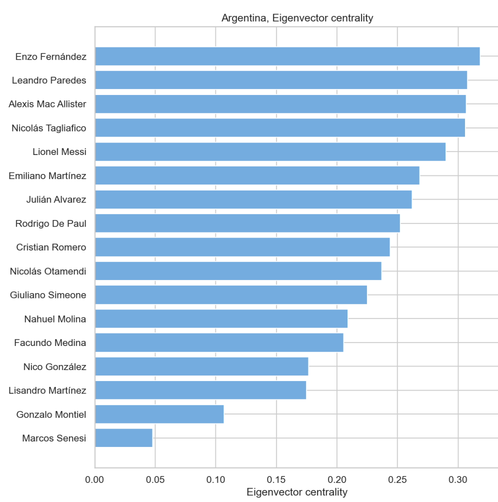

Fig. 12. Eigenvector centrality for Argentina. Enzo Fernández leads, consistent with his role as the structural anchor of Argentina's midfield.

D. Degree Distribution

The degree distributions confirm the tactical contrast between the two teams. Spain shows a relatively uniform distribution of passing volume across players, consistent with a rotational possession system where the ball moves fluidly through all lines. Argentina's distribution is more skewed, with Enzo Fernández and Leandro Paredes accounting for a disproportionate share of pass volume, suggesting a more centralized midfield structure with greater dependence on a smaller set of players. Unlike large-scale social networks, neither distribution follows a power law, which is expected given the bounded and tactically constrained nature of an 11-player passing system [4].

E. Temporal Analysis

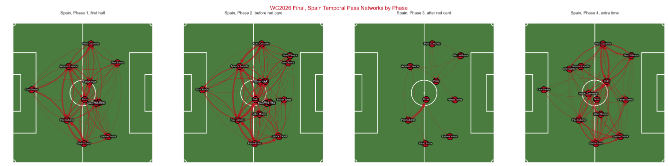

Fig. 13. Spain's pass network across the four match phases. Pass density remains consistently high through all phases.

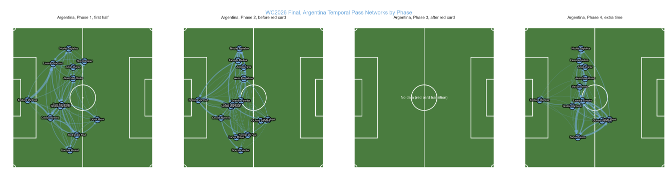

Fig. 14. Argentina's pass network across the four match phases. Phase 3 is empty following the red card dismissal.

Table III shows the evolution of both networks across the four match phases. Spain consistently outpaced Argentina in total passes and unique connections in every phase. The most revealing observation concerns Phase 3, the window immediately following Enzo Fernández's dismissal at expanded minute 97: Argentina recorded zero completed passes in this phase, indicating a complete breakdown in their ability to circulate the ball in the minutes immediately after the red card. In Phase 4, extra time, Argentina recovered some passing activity (95 passes, 50 connections) but remained well below Spain's output (188 passes, 77 connections), and their network density dropped relative to Phase 2, while Spain's remained stable.

TABLE III
TEMPORAL PHASE COMPARISON: SPAIN VS ARGENTINA

Phase	Team	Nodes	Edges	Passes	Density
Phase 1: First half	Spain	11	79	327	0.718
	Argentina	12	66	164	0.500
Phase 2: Before red card	Spain	15	94	263	0.448
	Argentina	14	75	118	0.412
Phase 3: After red card	Spain	9	13	14	0.181
	Argentina	0	0	0	0.000
Phase 4: Extra time	Spain	13	77	188	0.494
	Argentina	11	50	95	0.455

F. Community Detection

The Louvain algorithm identified three communities in each team's network, with modularity scores of 0.162 for Spain and 0.183 for Argentina.

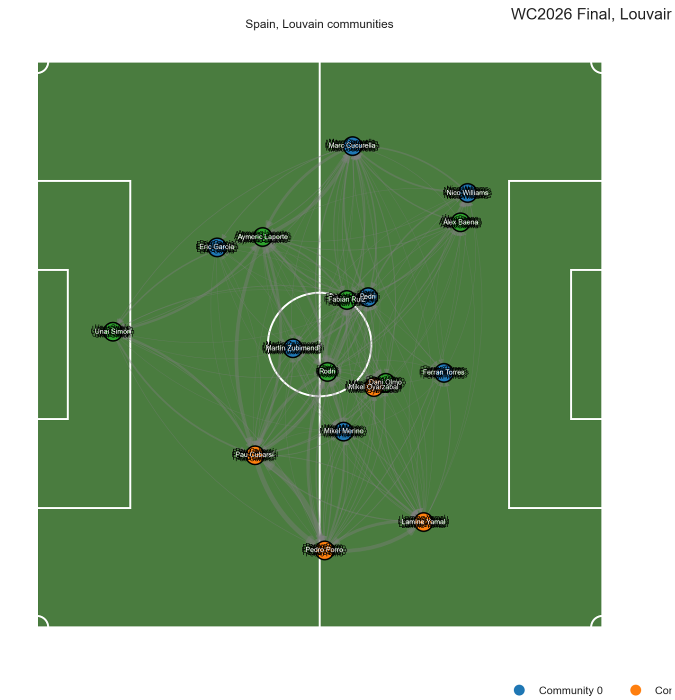

Fig. 15. Louvain communities for Spain. Three tactically coherent groups emerge from the passing structure.

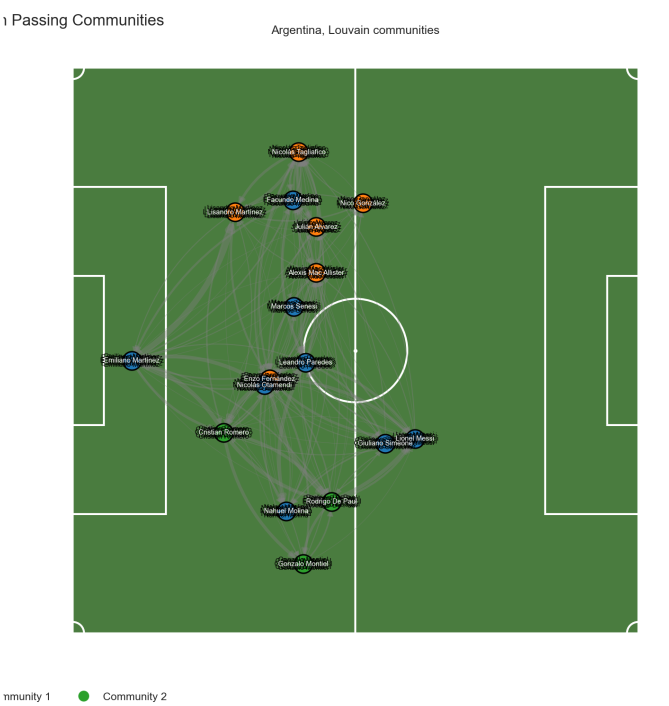

Fig. 16. Louvain communities for Argentina. Three groups reflect Argentina's more defensive and spatially compact structure.

Spain's three communities correspond to identifiable tactical units. The first community (Cucurella, Eric García, Zubimendi, Merino, Pedri, Nico Williams, Ferrán Torres) concentrated passing flow through the left and central channels, operating at an average pitch position of (56.3, 59.5). The internal cohesion of this group was directly responsible for creating the conditions that led to Ferrán Torres' winning goal: Cucurella's circular combinations with Pedri and Nico Williams repeatedly drew Argentina's attention to the left flank before Williams delivered the decisive assist to Torres in a central position.

The second community (Cubarśi, Porro, Lamine Yamal, Oyarzabal) applied pressure on Argentina's defensive line through a different mechanism. Argentina prioritized blocking Lamine Yamal, whom they treated as Spain's primary attacking threat, which freed space for Porro and Oyarzabal to operate and created additional passing lanes back to Cubarśi. The third community (Unai Simón, Laporte, Fabián Ruiz, Rodri, Dani Olmo, Álex Baena) functioned as the structural bridge connecting the other two groups, with Laporte and Rodri acting as the primary relay nodes for the lateral exchange between Cucurella on the left and Cubarśi in the center.

G. *Passing Behavior*

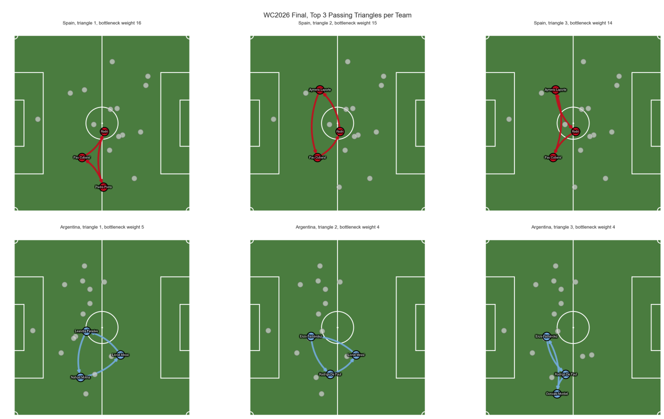

Fig. 17. Top three directed passing triangles for each team. Spain's triangles involve defensive and midfield players. Argentina's are centered on Enzo Fernández.

Spain recorded 341 unique directed passing triangles (cycles of the form $A \rightarrow B \rightarrow C \rightarrow A$), compared to Argentina's 267. The top triangle for Spain was Cubarśi $\rightarrow$ Rodri $\rightarrow$ Pedro Porro with a bottleneck weight of 16, followed by Laporte $\rightarrow$ Cubarśi $\rightarrow$ Rodri (15) and Laporte $\rightarrow$ Rodri $\rightarrow$ Cubarśi (14). The consistent presence of Cubarśi, Rodri, Laporte, and Porro across the top triangles confirms that Spain's triangular passing combinations were concentrated in the defensive and central midfield zones, used primarily as a ball retention mechanism to build play from the back rather than as a direct attacking tool [5].

Argentina's top triangle was Paredes $\rightarrow$ Molina $\rightarrow$ Messi with a bottleneck weight of only 5, and Enzo Fernández appeared in the majority of Argentina's top-10 triangles. The bottleneck weights are substantially lower than Spain's, reflecting both fewer total passes and less repetition of specific triangular combinations. The dependence of Argentina's triangle structure on Enzo Fernández further underscores the structural damage caused by his dismissal: not only did Argentina lose a player, they lost the node around which most of their triangular combinations were organized.

V. Conclusions

The graph-theoretic analysis of the 2026 FIFA World Cup final provides quantitative support for the observation that Spain's victory was the product of structural superiority rather than circumstance. Spain's passing network was denser, more distributed, and more tactically organized than Argentina's across all four match phases. The 792 completed passes Spain recorded against Argentina's 377 reflect a sustained and system-level difference in ball control, not merely individual quality.

The centrality analysis revealed that Spain's most structurally important player was not a midfielder or a forward but left back Marc Cucurella, who led in betweenness, eigenvector, and in-degree centrality. This finding reflects a deliberate tactical choice to route Spain's ball circulation through the left flank, a pattern that ultimately created the conditions for the winning goal. Pau Cubarśi's leadership in PageRank confirms that the defensive line was the origin point of Spain's possession sequences, with Cubarśi serving as the lateral relay between Spain's two flanks.

Argentina's Enzo Fernández emerged as the single most structurally critical player on his team, leading in four of the five centrality measures computed. His dismissal at expanded minute 97 did not merely reduce Argentina to ten players; it removed the node around which their triangular combinations and passing connectivity were organized, as evidenced by the zero completed passes Argentina recorded in the immediately following phase. The community detection results further confirm that Spain's three tactical groups were internally coherent and contributed distinct tactical functions to the match outcome, while Argentina's communities reflected a more defensive and spatially compressed structure.

Taken together, these results demonstrate that social network analysis is a rigorous and practically applicable framework for understanding football matches beyond subjective observation, with direct utility for coaching staff, tactical analysts, and sports data scientists.

VI. Acknowledgments

I would like to express my gratitude to my coordinator and professor Didier Omar Gamboa Angulo for the time and knowledge shared throughout the Social Networks Analysis course, as well as to his teaching assistant José Francisco Pérez Alcocer for providing valuable insight into football strategy and tactics. I also extend my appreciation to PhD. Soledad Pech Couoh, my Unsupervised Learning professor, whose lectures on clustering methods directly informed the community detection approach applied in this project. Finally, I would like to thank my sister Belinda De Aquino Castellanos for teaching me everything I know about football and for sparking my appreciation for the strategic depth of the game, particularly in one of my favorite national teams: Spain.

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